AUTOMATED VEHICLE CLASSIFICATION & COUNTING

AVC&C | Group G25 | University of Bradford

FAIR REPORT

AI & Generative AI Disclosure | Responsible Use Declaration

1. Purpose of This Document

This FAIR Report (Findable, Accountable, Informed, Responsible) discloses all use of Artificial Intelligence and Generative AI tools during the development of the AVC&C project by Group G25. This document is produced in accordance with the University of Bradford's academic integrity policy and the expectations of COS6032-E Industrial AI Project module assessment.

The document covers: which AI tools were used, by whom, for what purpose, what was produced by AI versus by the team, and how AI-generated content was reviewed and validated before inclusion in any deliverable.

2. AI Tools Declared

AI Tool	Type	Used By	Purpose
Claude (Anthropic)	Generative AI — LLM	Ayush Acharya	Code debugging, documentation, design guidance
YOLO11n (Ultralytics)	AI — Object Detection	Ayush Acharya	Core vehicle detection model (primary system)
DeepSORT	AI — Multi-Object Tracking	Ayush Acharya	Vehicle tracking across frames
Roboflow	AI-assisted annotation	Pratham Patel	Dataset labelling assistance and management
Canva (AI features)	Generative AI — Design	Aman Gill	Exhibition poster layout suggestions

3. Generative AI Usage Detail — Claude (Anthropic)

Claude AI (claude.ai) was used by Ayush Acharya as a development and documentation assistant throughout the project. All usage is declared below in full, categorised by activity type.

3.1 Code Debugging & Development Support

Task	AI Role	Team Role
vehicle_counter.py — double-counting bug fix	Suggested centroid history approach	Ayush implemented, tested, and validated fix
data.yaml path error (spaces in folder name)	Suggested raw string r'...' fix	Ayush applied and verified across all scripts

DeepSORT integration debugging	Advised on track ID management	Ayush implemented full tracking pipeline
Streamlit dashboard auto-refresh logic	Suggested st.rerun() pattern	Aman implemented and customised dashboard
YOLO11n training script (train.py)	Reviewed config parameters	Ayush wrote, ran, and validated full training

3.2 Documentation & Writing Support

Document	AI Contribution	Team Contribution
Agile Artefacts document	Structural guidance, formatting	All content, data, and sprint logs written by team
Testing & Validation Report	Document structure and table design	All test results, data, and analysis by team
README.md	Formatting and structure suggestions	All technical content written by Ayush
Exhibition Poster content	Copy editing and layout advice	All technical content and design by team
This FAIR Report	Document generation (declared here)	All declared facts provided and verified by team

3.3 Design Guidance

- **Dashboard UI:** Claude provided advice on Streamlit layout patterns and colour-coded bounding box implementation. Final design decisions and implementation by Aman Gill.
- **Exhibition boards:** Claude advised on information architecture for the A1 poster. All visual design executed in Canva by Aman Gill.
- **System architecture:** Claude reviewed proposed YOLO + DeepSORT + Streamlit pipeline design. Architecture decisions made by Group G25 with Dr Panesar's supervision.

4. What AI Did NOT Do

The following activities were performed entirely by Group G25 without AI generation. This section is included to clearly establish the boundaries of AI assistance and confirm the academic integrity of the submitted work.

Activity	Performed By
Dataset sourcing and weather condition selection	Pratham Patel — via Roboflow
11,582 image annotation and labelling	Pratham Patel (~25 hours manual effort)
70/20/10 train/val/test split strategy	Ayush Acharya (prepare_dataset.py)
YOLO11n model training (50 epochs, ~3 hours)	Ayush Acharya — CPU training run
UA-DETRAC benchmark evaluation	Pratham Patel — Sprint 7
Bradford manual ground truth count (48 vehicles)	Ayush Acharya — manual verification
GDPR compliance audit	Ayush Acharya — Risk Owner R1
Client requirements gathering (Bradford Council)	All team — Sprint 1 meeting with Yunus Mayat
System architecture design decisions	Group G25 with Dr Panesar supervision
Sprint planning, retrospectives, risk reviews	All team — Ayush as Scrum Master
All model weights and performance results	Generated by team training pipeline

5. AI Output Validation Process

All AI-generated or AI-assisted content was reviewed and validated before inclusion in any project deliverable. The following process was applied consistently:

#	Step	Description
1	Critical Review	All AI suggestions reviewed by the team member responsible for that component before implementation
2	Empirical Testing	Code suggestions tested against actual data — if performance did not match, AI suggestion was discarded
3	Cross-Verification	At least one other team member verified significant changes to shared code components
4	Supervisor Review	Dr Panesar reviewed system outputs and documentation at sprint reviews — flagged issues were resolved by team
5	No Fabrication Rule	No AI-generated data, statistics, or test results were used. All metrics are from actual model runs and tests

6. YOLO11n & DeepSORT — AI Model Transparency

The AVC&C system is itself an AI system. The following transparency information is provided about the core AI components:

6.1 YOLO11n (Object Detection Model)

- **Model Source:** Ultralytics YOLO11n — open-source, Apache 2.0 licensed
- **Pretrained On:** COCO dataset (Common Objects in Context) — 80 object classes
- **Fine-tuned On:** Group G25 custom dataset — 11,582 images, 5 weather conditions
- **Classes Used:** Car (COCO class 2), Bus (class 5), Truck (class 7) — filtered from 80
- **Training Transparency:** All training hyperparameters documented in Section 3.1 of Testing & Validation Report
- **Performance Validated:** mAP@0.5: 91.4% — verified on held-out test set (1,713 images) and UA-DETRAC benchmark
- **Limitations:** Motorcycles and bicycles excluded from v1.0. Performance on night-time footage not tested.

6.2 DeepSORT (Multi-Object Tracking)

- **Algorithm:** Deep SORT — Simple Online and Realtime Tracking with a Deep Association Metric
- **Purpose:** Assigns persistent track IDs to detected vehicles across video frames
- **Privacy Role:** Track IDs reset on each session restart — no persistent individual tracking
- **Counting Logic:** Virtual trip-line at y=300 — counts only confirmed directional centroid crossings

7. Responsible AI Principles Applied

Group G25 applied the following responsible AI principles throughout the project, aligned with Bradford Council's requirements and UK AI governance frameworks:

Principle	How Applied	Evidence
Transparency	Declared all AI tools used in this document	This FAIR Report
Privacy by Design	No personal data captured — counts only	GDPR audit — 100% pass
Human Oversight	All AI suggestions reviewed before use	Manual ground truth: 48/48 match
Explainability	Dashboard shows class-level data with weather filter	app.py — Streamlit dashboard
Fairness	Tested across 5 weather conditions equally	Per-weather results table
Accountability	Clear team ownership per deliverable	Agile Artefacts — roles table
Academic Integrity	All AI use declared; no AI-generated data	This document

8. Declaration

We, the undersigned members of Group G25, confirm that:

- All AI tool usage during this project is fully and accurately declared in this document
- No AI-generated data, statistics, benchmark results, or test outputs have been presented as our own empirical findings
- All model training, dataset preparation, system architecture decisions, and validation were performed by Group G25
- AI tools were used as assistants to support our work, not to replace our academic and technical effort
- This document has been reviewed for accuracy by all three team members

Name	Role	Signature / Date
Ayush Acharya	AI/ML Lead & Scrum Master	April 17, 2026
Pratham Patel	Data & Testing Lead	April 17, 2026
Aman Gill	Dashboard & Documentation Lead	April 17, 2026